

Appendix A

Simulation Pseudocode for an ABM of Brokerage in Matching Markets

1 State, Notation, and Matching Function

Blue notation, equations, and algorithm lines are diagnostics-only measures. They are recorded for $\mathcal{M}$ or downstream figures, exploration scripts, and analyses; they do not enter transition, choice, learning, or matching rules.

For any matrix Z , Z_ℓ denotes column ℓ and $Z_{.L}$ denotes the submatrix with ordered column-index set L . For a vector z , z_L denotes the corresponding subvector.

Table 1: Cross-routine symbols and dimensions.

Symbol	Dimension/domain	Definition
<i>Core</i>		
N	scalar	Number of agents, with agent set $[N] = \{1, \dots, N\}$.
ϑ	parameter vector	Simulation parameter vector.
seed	integer	Random seed.
S^t	state tuple	Mutable simulation state at end of period t .
T	integer	Number of simulation periods.
$\mathcal{M} = [m^1; \dots; m^T]$	$\mathbb{R}^{T \times m}$	Measure table; row $m^t = \mathcal{M}_t \in \mathbb{R}^m$.
<i>Parameters</i>		
d	scalar	Agent type dimension.
k_G	scalar	Watts-Strogatz graph degree.
ω	scalar	Satisfaction EWMA weight.
$\mathbf{L}$	learning function	NN (default) or Ridge.
$\mathbf{S}_b$	broker service	Full (default), assessment-only, or access-only.
λ_a, λ_b	positive scalars	Agent and broker Ridge slope penalties; calibrated experiment values are $\lambda_a = 0.003$ and $\lambda_b = 0.001$.
a_b	broker Ridge variant	Pair, size-matched, single-principal, or additive.
$E_{\text{init},a}, E_{\text{init},b}$	scalars	Agent and broker initial training steps; both are 100 in the reporting simulations.
$E_{\text{steps},a}, E_{\text{steps},b}$	scalars	Agent and broker minimum training steps per fitting call; both are 50 in the reporting simulations.
$\eta_{\text{r},a}, \eta_{\text{r},b}$	scalars	Agent and broker Adam learning rates; reporting values are 0.003 and 0.03, respectively.
W_p	scalar	Training-window horizon, in periods.
M	scalar	Per-call training observation cap.
<i>Agent types, networks, and output</i>		
$G^t = (V, \mathcal{E}_G^t)$	graph	Undirected graph, $V = [N] \cup \{N + 1\}$; node $N + 1$ is the broker.

Symbol	Dimension/domain	Definition
$X^t = [x_1^t, \dots, x_N^t]$	$\mathbb{R}^{d \times N}$	Agent type matrix; $x_i^t = X_{\cdot i}^t \in \mathbb{R}^d$.
γ	$[0, 1] \rightarrow \mathbb{R}^d$	Agent type curve for initial and entrant draws.
q_{ij}^t	scalar	Realized output for match (i, j) .
<i>Match-output function</i>		
Q	scalar	Output offset.
ρ	scalar	General-quality share of the two components' summed marginal variances.
δ	scalar	Strength of the pair boundary in match-specific complementarity.
σ_ε	scalar	Output-noise standard deviation.
c	$\mathbb{R}^d$	Ideal agent type.
A	$\mathbb{R}^{d \times d}$	Symmetric positive definite complementarity matrix.
B	$\mathbb{R}^{d \times d}$	Symmetric boundary matrix.
<i>Calibration and costs</i>		
$\bar{q}_{\text{cal}}$	scalar	Calibration mean.
r	scalar	Reservation output.
ϕ	scalar	Broker fee.
c_s	scalar	Self-search cost.
<i>Learning and predictions</i>		
d_b	integer	Broker symmetric pair-feature dimension, $d + d(d + 1)/2$.
$\psi_b(x_i, x_j)$	$\mathbb{R}^{d_b}$	Symmetric broker pair feature vector.
$\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$	$\Xi_i^t \in \mathbb{R}^{d \times n_i^t}, y_i^t \in \mathbb{R}^{n_i^t}$	Agent i history; observation ℓ is partner agent type $\Xi_{i,\ell}^t$ and output $y_{i\ell}^t$.
$\mathcal{H}_b^t = (X_{b,1}^t, X_{b,2}^t, y_b^t, n_b^t)$	$X_{b,1}^t, X_{b,2}^t \in \mathbb{R}^{d \times n_b^t}, y_b^t \in \mathbb{R}^{n_b^t}$	Broker history; observation ℓ stores the two party types and output $y_{b\ell}^t$.
Θ_i^t	predictor parameters	Agent i NN weights or Ridge coefficients.
Θ_b^t	predictor parameters	Broker NN weights or Ridge coefficients.
$\mathcal{O}_i^t, \mathcal{O}_b^t$	Adam states	Persistent NN optimizer moments and step counters; unused by Ridge.
Δ_i^t, Δ_b^t	nonnegative integers	Observations added since the learner's previous training call.
$\mathbf{p}_i^t, \mathbf{p}_b^t$	integer sequences	Cumulative history counts at the end of learner initialization (period 0) and each later completed period.
$v_b^t(i, j)$	scalar	Symmetric broker prediction for unordered pair $\{i, j\}$.
$a_{i \rightarrow j}^t$	scalar	Agent i 's assessment of candidate j .
$w_{j \leftarrow i}^t$	scalar	Receiver acceptance score.
$\bar{q}_i^t$	vector	Agent i 's partner-indexed empirical means.
C_i^t	vector	Agent i 's partner-indexed counts.
<i>Period state and market objects</i>		
s_i^t	$\mathbb{R}^2$	Satisfaction scores for self and broker channels.
χ_i^t	binary	Indicator that agent i has ever used the broker.
M_i^t	set	Agent i current-period counterparties.
z_i^t	channel	Channel chosen by demander i .
d_i^t	integer	Current-period demand.
D^t	subset of $[N]$	Current broker clients, $\{i : d_i^t > 0, z_i^t = \text{broker}\}$.
$\mathcal{A}_b^t$	subset of $[N]$	Broker access set, $\text{Roster}^t \cup D^t$.
Roster^t	subset of $[N]$	Standing roster.

Symbol	Dimension/domain	Definition
rep_b^t	scalar	Broker reputation.
D_+^t	subset of $[N]$	Broker clients with positive residual offer quota.
$\mathcal{D}^t$	list	Demand records (i, z_i^t, d_i^t) with $d_i^t > 0$.
$\mathcal{R}^t$	list	Accepted records $(i, j, z, q, \hat{q}, o_1, o_2)$.
rng^t	RNG state	State of the simulation random-number stream.

The mutable simulation state at the end of period t is

$$S^t = \left(\vartheta, \text{rng}^t, \gamma, c, A, B, \bar{q}_{\text{cal}}, r, \phi, c_s, X^t, G^t, \right. \\ \left. \{\mathcal{H}_i^t, \Theta_i^t, \mathcal{O}_i^t, \Delta_i^t, \mathbf{p}_i^t, \bar{q}_i^t, C_i^t, s_i^t, \chi_i^t, M_i^t\}_{i=1}^N, \right. \\ \left. \mathcal{H}_b^t, \Theta_b^t, \mathcal{O}_b^t, \Delta_b^t, \mathbf{p}_b^t, \text{Roster}^t, D^t, \text{rep}_b^t \right).$$

This tuple includes fixed realized model objects and all variables needed for later transitions. Implementation-only scratch buffers and diagnostic-only counters are omitted.

For the fixed reference moments, define

$$U_{ij} = \frac{x_i^\top c + x_j^\top c}{2}, \quad V_{ij} = x_i^\top A x_j, \quad H_{ij} = \text{sign}(x_i^\top B x_j) V_{ij}, \\ \tilde{U}_{ij} = \frac{U_{ij} - \mu_U}{\sigma_U}, \quad \tilde{W}_{ij}(\delta) = \frac{V_{ij} + \delta H_{ij} - (\mu_V + \delta \mu_H)}{\sqrt{\sigma_V^2 + 2\delta \text{Cov}(V, H) + \delta^2 \sigma_H^2}}.$$

With $\kappa(\delta) = \text{Corr}(U, W(\delta))$, the systematic output component is

$$f(x_i, x_j) = \frac{\sqrt{\rho} \tilde{U}_{ij} + \sqrt{1-\rho} \tilde{W}_{ij}(\delta)}{\sqrt{1 + 2\sqrt{\rho(1-\rho)} \kappa(\delta)}}.$$

Expected match output is $Q + f(x_i, x_j)$. A completed match produces realized output

$$q_{ij} = Q + f(x_i, x_j) + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2).$$

2 Top-Level Simulation

Algorithm 1 Brokerage Simulation

Require: Parameters ϑ and seed.

Ensure: Final state S^T and **measure table** $\mathcal{M} \in \mathbb{R}^{T \times m}$.

- 1: $S^0 \leftarrow \text{INITIALIZE}(\vartheta, \text{seed})$
 - 2: **for** $t = 1, \dots, T$ **do**
 - 3: $(S^t, \mathcal{R}^t, Y^t) \leftarrow \text{PERIODUPDATE}(S^{t-1}, t, \vartheta)$
 - 4: $\mathcal{M}_t \leftarrow Y^t$
 - 5: **return** $(S^T, \mathcal{M})$
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3 Learning and Prediction

Let $\mathbf{L} \in \{\text{NN}, \text{Ridge}\}$ select the learning function. The default is $\mathbf{L} = \text{NN}$. Both branches use the same histories, period window, and observation cap. Inputs remain on their raw DGP scale.

For any one-hidden-layer predictor with input dimension p_{in} and hidden width h ,

$$\text{NN}_{\Theta}(\xi) = w_2^\top \text{ReLU}(W_1 \xi + b_1) + b_2, \quad \Theta = (W_1, b_1, w_2, b_2),$$

where $\xi \in \mathbb{R}^{p_{\text{in}}}$, $W_1 \in \mathbb{R}^{h \times p_{\text{in}}}$, $b_1, w_2 \in \mathbb{R}^h$, and $b_2 \in \mathbb{R}$.

$$d_b = d + \frac{d(d+1)}{2}, \quad \psi_b(x_i, x_j) = \left[x_i + x_j; \text{vech}\left(\frac{x_i x_j^\top + x_j x_i^\top}{2}\right) \right].$$

Agent networks use $(p_{\text{in}}, h) = (d, 2d)$; the broker network uses $(p_{\text{in}}, h) = (d_b, 8d)$.

$$\psi_b(x_i, x_j) = \psi_b(x_j, x_i)$$

so broker histories produce one training column per observed unordered pair.

Local variables for training

Symbol	Dimension/domain	Definition
p_{in}, h	integers	Predictor input dimension and hidden width.
ξ	$\mathbb{R}^{p_{\text{in}}}$	Predictor input vector.
$\mathbf{L}$	learning function	NN or Ridge.
a_b	broker Ridge variant	Pair, size-matched, single-principal, or additive.
Θ	predictor parameters	NN parameters (W_1, b_1, w_2, b_2) or Ridge parameters (α, β) .
W_1, b_1, w_2, b_2	matrix, vectors, scalar	One-hidden-layer MLP weights and biases.
d_b	integer	Broker symmetric pair-feature dimension.
ψ_b	function	Symmetric broker pair-feature map.
$\Xi_{\text{tr}}, y_{\text{tr}}$	$\mathbb{R}^{p_{\text{in}} \times n}, \mathbb{R}^n$	Generic training inputs and targets.
$\mathcal{L}$	scalar	Full-batch MSE objective.
λ	positive scalar	Learner-specific Ridge slope penalty; intercepts are not penalized.
e	integer	Update-step index.
(m, v, τ)	Adam state	Persistent Adam moments and step counter; $\mathcal{O}_i, \mathcal{O}_b$ are the agent and broker instances.
$\beta_1, \beta_2, \epsilon$	scalars	Adam constants $(0.9, 0.999, 10^{-8})$.
n_i, n_b	integers	Current values of history lengths n_i^t and n_b^t .
L_i, L_b	ordered index sets	Training-window indices from the most recent W_p completed simulation periods, with period 0 included only until W_p later periods exist; subsampled evenly to at most M .
Δ_i, Δ_b	integers	New observations since last training call.
E_i, E_b	integers	Agent and broker NN update-step counts.
$\Xi_{i,\text{tr}}, y_{i,\text{tr}}$	$\mathbb{R}^{d \times L_i }, \mathbb{R}^{ L_i }$	Agent i training view of $\mathcal{H}_i^t$.
$Z_{b,\text{tr}}, y_{b,\text{tr}}$	matrix, $\mathbb{R}^{ L_b }$	Broker fitting view using the selected input map.
r_ℓ	$\{1, 2\}$	Endpoint retained once when broker observation ℓ is recorded for the single-principal ablation.

Algorithm 2 TrainNN

Require: Network parameters Θ , inputs $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}} \times n}$, targets $y_{\text{tr}} \in \mathbb{R}^n$, steps E , learning rate η_{lr} , persistent Adam state (m, v, τ) .

- 1: **for** $e = 1, \dots, E$ **do**
- 2: $g \leftarrow \nabla_{\Theta} \mathcal{L}(\Theta)$, $\mathcal{L}(\Theta) = n^{-1} \sum_{\ell=1}^n (\text{NN}_{\Theta}(\Xi_{\text{tr}, \cdot \ell}) - y_{\text{tr}, \ell})^2$
- 3: $\tau \leftarrow \tau + 1$
- 4: $m \leftarrow \beta_1 m + (1 - \beta_1)g$, $v \leftarrow \beta_2 v + (1 - \beta_2)g \odot g$
- 5: $\Theta \leftarrow \Theta - \eta_{\text{lr}} \frac{m/(1 - \beta_1^{\tau})}{\sqrt{v/(1 - \beta_2^{\tau})} + \epsilon}$
- 6: **return** $\Theta, (m, v, \tau)$

Algorithm 3 FitPredictor

Require: Learning function $\mathbf{L}$, learner $c \in \{a, b\}$, parameters Θ , raw inputs $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}} \times n}$, targets $y_{\text{tr}} \in \mathbb{R}^n$, learner-specific Ridge penalty λ_c , and fitting parameters.

- 1: **if** $\mathbf{L} = \text{NN}$ **then**
- 2: $E \leftarrow \max\{E_{\text{steps}, c}, \lceil E_{\text{init}, c} \Delta / n_{\text{total}} \rceil\}$.
- 3: $(\Theta, \mathcal{O}) \leftarrow \text{TRAINNN}(\Theta, \Xi_{\text{tr}}, y_{\text{tr}}, E, \eta_{\text{lr}, c}, \mathcal{O})$.
- 4: **else**
- 5: $(\alpha, \beta) \leftarrow \arg \min_{a, b} \left\{ n^{-1} \sum_{\ell=1}^n (a + b^{\top} \Xi_{\text{tr}, \cdot \ell} - y_{\text{tr}, \ell})^2 + \lambda_c \|b\|_2^2 \right\}$.
- 6: $\Theta \leftarrow (\alpha, \beta)$.
- 7: **return** Θ and, for NN, $\mathcal{O}$.

The agent input is always the prospective partner type. The broker Ridge training input for observation ℓ is

$$z_{b\ell} = \begin{cases} \psi_b(x_{1\ell}, x_{2\ell}), & a_b \in \{\text{pair, size-matched}\}, \\ x_{r_{\ell}, \ell}, & a_b = \text{single-principal}, \\ x_{1\ell} + x_{2\ell}, & a_b = \text{additive}. \end{cases}$$

For size-matched history, uniformly sample without replacement from the broker's windowed and capped observations. Its sample size is the smaller of the broker sample size and the nearest-integer median agent sample size among agents with nonempty histories, with half-integers rounded up.

Prediction is modular:

$$\text{PREDICTAGENT}(i, x_j) = \begin{cases} \text{NN}_{\Theta_i}(x_j), & \mathbf{L} = \text{NN}, \\ \alpha_i + \beta_i^{\top} x_j, & \mathbf{L} = \text{Ridge}, \end{cases}$$

and

$$\text{PREDICTBROKER}(i, j) = \begin{cases} \text{NN}_{\Theta_b}(\psi_b(x_i, x_j)), & \mathbf{L} = \text{NN}, \\ \alpha_b + \gamma_b^{\top} \psi_b(x_i, x_j), & \mathbf{L} = \text{Ridge}, a_b \in \{\text{pair, size-matched}\}, \\ \alpha_b + \gamma_b^{\top} (x_i + x_j), & \mathbf{L} = \text{Ridge}, a_b = \text{additive}, \\ g_b(x_i) + g_b(x_j) - \bar{y}_{b, \text{tr}}, & \mathbf{L} = \text{Ridge}, a_b = \text{single-principal}, \end{cases}$$

where $g_b(x) = \alpha_b + \gamma_b^{\top} x$. Before any Ridge fit, $\alpha = Q$, slopes are zero, and $\bar{y}_{b, \text{tr}} = Q$.

Within period-level routines, state variables without a superscript refer to their current-period values.

Algorithm 4 TrainPredictors

Require: State S , period t , learning function L , and fitting parameters.

- 1: **for** $i = 1, \dots, N$ in parallel **do**
 - 2: $n_i \leftarrow n_i^t$, $\Delta_i \leftarrow$ observations added to $\mathcal{H}_i$ since its last training call.
 - 3: **if** $n_i > 0$, $\Delta_i > 0$, and $i \bmod 2 = t \bmod 2$ **then**
 - 4: Let $P_i \leftarrow |\mathbf{p}_i|$ and $b_i \leftarrow 0$ if $P_i \leq W_p$, otherwise $b_i \leftarrow (\mathbf{p}_i)_{P_i - W_p}$; set $L_i \leftarrow \{b_i + 1, \dots, n_i\}$.
 If $|L_i| > M$, keep M spaced indices across L_i , including its first and last indices (or only its last index when $M = 1$).
 - 5: $\Xi_{i,\text{tr}} \leftarrow (\Xi_i^t)_{L_i}$, $y_{i,\text{tr}} \leftarrow (y_i^t)_{L_i}$.
 - 6: $\Theta_i \leftarrow \text{FITPREDICTOR}(L, a, \Theta_i, \Xi_{i,\text{tr}}, y_{i,\text{tr}}, \lambda_a, \Delta_i, n_i, \vartheta)$; $\Delta_i \leftarrow 0$.
 - 7: $n_b \leftarrow n_b^t$, $\Delta_b \leftarrow$ broker observations since last training.
 - 8: **if** $n_b > 0$ and $\Delta_b > 0$ **then**
 - 9: Let $P_b \leftarrow |\mathbf{p}_b|$ and $b_b \leftarrow 0$ if $P_b \leq W_p$, otherwise $b_b \leftarrow (\mathbf{p}_b)_{P_b - W_p}$; set $L_b \leftarrow \{b_b + 1, \dots, n_b\}$.
 If $|L_b| > M$, keep M spaced indices across L_b , including its first and last indices (or only its last index when $M = 1$).
 - 10: If $L = \text{Ridge}$ and $a_b = \text{size-matched}$, replace L_b by the uniform size-matched sample defined above.
 - 11: Construct $Z_{b,\text{tr}} = [z_{b\ell}]_{\ell \in L_b}$ and $y_{b,\text{tr}} \leftarrow (y_b^t)_{L_b}$.
 - 12: $\Theta_b \leftarrow \text{FITPREDICTOR}(L, b, \Theta_b, Z_{b,\text{tr}}, y_{b,\text{tr}}, \lambda_b, \Delta_b, n_b, \vartheta)$; $\Delta_b \leftarrow 0$.
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4 Initialization

Calibration draws random pairs (i_ℓ, j_ℓ) and sets

$$\bar{q}_{\text{cal}} = \frac{1}{10,000} \sum_{\ell=1}^{10,000} \{Q + f(x_{i_\ell}, x_{j_\ell})\},$$

$$r = f_r \bar{q}_{\text{cal}}, \quad \phi = c_s = \lambda_c \bar{q}_{\text{cal}}.$$

Local variables for Initialize

Symbol	Dimension/domain	Definition
λ_c	scalar	Cost rate used to set ϕ and c_s .
f_r	scalar	Reservation coefficient; sets $r = f_r \bar{q}_{\text{cal}}$ (default 0.60).
(i_ℓ, j_ℓ)	pair in $[N]^2$	Random calibration pair.
d_γ	integer	Number of active curve dimensions.
p_{rewire}	scalar	Watts-Strogatz rewiring probability.
ν_ℓ	integer	Frequency for active curve coordinate ℓ .
ψ_ℓ	scalar	Phase for active curve coordinate ℓ .
A_0	$\mathbb{R}^{d \times d}$	Raw matrix used to construct A .
Σ_x	$\mathbb{R}^{d \times d}$	Empirical agent type second moment.
B_0	$\mathbb{R}^{d \times d}$	Raw zero-trace symmetric boundary matrix.
B_{raw}	$\mathbb{R}^{d \times d}$	Orthogonalized pre-normalization boundary matrix.
$Z_{b,\text{seed}}, y_{b,\text{seed}}$	matrix, $\mathbb{R}^{n_b^0}$	Broker seed fitting batch using the selected input map.

Local variables for InitializeAgent

Symbol	Dimension/domain	Definition
σ_x	scalar	Agent type noise scale.
u_i	scalar	Agent i curve position.
ϵ_i	$\mathbb{R}^d$	Agent type noise for agent i .
π_{ij}^{ent}	scalar	Entrant attachment probability.
$\mathcal{V}_i^{\text{ent}}$	subset of $[N] \setminus \{i\}$	Entrant attachment set.

Algorithm 5 InitializeAgent

Require: State S , agent slot i , period counter t , parameters ϑ .

Ensure: Agent slot i is initialized in S .

- 1: Draw $u_i \sim \text{Unif}[0, 1]$ and $\epsilon_i \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$.
 - 2: $x_i^t \leftarrow (\gamma(u_i) + \epsilon_i) / \|\gamma(u_i) + \epsilon_i\|_2$.
 - 3: Reset $\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$ to zero observations, $\bar{q}_i^t$, C_i^t , χ_i^t , M_i^t , and Δ_i^t .
 - 4: Initialize the selected predictor to the constant Q : for NN, draw W_{1i}^t with He initialization and set $b_{1i}^t = 0$, $w_{2i}^t = 0$, $b_{2i}^t = Q$, $\mathcal{O}_i^t = 0$; for Ridge, set $\alpha_i^t = Q$ and $\beta_i^t = 0$.
 - 5: **if** $t = 0$ **then**
 - 6: $s_i^t \leftarrow (0, 0)$.
 - 7: **else**
 - 8: Remove i from Roster^t , D^t , and M_j^t for all j ; delete all incident edges in G^t ; set $\bar{q}_{ji}^t, C_{ji}^t \leftarrow 0$ for all $j \neq i$.
 - 9: Sample $\mathcal{V}_i^{\text{ent}} \subset [N] \setminus \{i\}$ without replacement, $|\mathcal{V}_i^{\text{ent}}| = \min(\lfloor k_G/2 \rfloor, N-1)$, with

$$\pi_{ij}^{\text{ent}} \propto \exp\{-\|x_i^t - x_j^t\|_2^2\}.$$
 - 10: Add edges (i, j) to G^t for all $j \in \mathcal{V}_i^{\text{ent}}$.
 - 11: $s_{i,\text{self}}^t \leftarrow \begin{cases} |\mathcal{V}_i^{\text{ent}}|^{-1} \sum_{j \in \mathcal{V}_i^{\text{ent}}} s_{j,\text{self}}^t, & \mathcal{V}_i^{\text{ent}} \neq \emptyset, \\ 0, & \mathcal{V}_i^{\text{ent}} = \emptyset. \end{cases}$
 - 12: $s_{i,\text{broker}}^t \leftarrow \text{rep}_b^t$.
 - 13: Set $\mathbf{p}_i^t \leftarrow (0)$, treating entrant initialization as period 0 of its tenure.
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Algorithm 6 Initialize

Require: Parameters ϑ and seed.

Ensure: Initial state S^0 .

- 1: Draw curve frequencies $\nu_\ell \sim \text{Unif}\{1, \dots, 5\}$ and phases $\psi_\ell \sim \text{Unif}[0, 2\pi)$ for $\ell = 1, \dots, d_\gamma$, defining γ .
- 2: Allocate empty agent and broker containers in S^0 and store γ .
- 3: **for** $i = 1, \dots, N$ **do**
- 4: INITIALIZEAGENT($S^0, i, 0, \vartheta$).
- 5: Draw $u_c \sim \text{Unif}[0, 1]$ and $\epsilon_c \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$; set $c \leftarrow [\gamma(u_c) + \epsilon_c] / \|\gamma(u_c) + \epsilon_c\|_2$.
- 6: Draw $A_0 \in \mathbb{R}^{d \times d}$ iid standard normal and set $A \leftarrow A_0^\top A_0 \cdot d / \text{tr}(A_0^\top A_0)$.
- 7: $\Sigma_x \leftarrow N^{-1} \sum_i x_i^0 (x_i^0)^\top$.
- 8: Draw symmetric zero-trace B_0 and set

$$B_{\text{raw}} \leftarrow B_0 - \frac{\text{tr}(\Sigma_x B_0 \Sigma_x A)}{\text{tr}(\Sigma_x A \Sigma_x A)} A, \quad B \leftarrow B_{\text{raw}} / \|B_{\text{raw}}\|_F.$$

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9: Calibrate  $(\bar{q}_{\text{cal}}, r, \phi, c_s)$ .
10: Build  $G^0$  as a Watts-Strogatz graph on  $[N]$  with degree  $k_G$  and rewiring probability  $p_{\text{rewire}}$ ; add isolated broker node  $N + 1$ .
11: Initialize  $\mathcal{H}_b^0$  and the selected broker predictor to the constant  $Q$  using the corresponding NN or Ridge initialization above; set  $\text{rep}_b^0 = 0$ .
12: Sample  $\text{Roster}^0 \subset [N]$  uniformly with  $|\text{Roster}^0| = \lceil \alpha_R N \rceil$ ; add broker edges  $(i, N + 1)$  for all roster members.
13: for each unordered non-broker edge  $\{i, j\} \in \mathcal{E}_G^0$  do
14:   Draw  $q_{ij}$  once; append  $(x_j^0, q_{ij})$  to  $\mathcal{H}_i^0$  and  $(x_i^0, q_{ij})$  to  $\mathcal{H}_j^0$ ; update  $\bar{q}_{ij}^0, C_{ij}^0, \bar{q}_{ji}^0, C_{ji}^0$ .
15: Let  $\mathcal{B}_{\text{seed}} \leftarrow \{\{i, j\} \in \mathcal{E}_G^0 : i, j \in \text{Roster}^0\}$ ; shuffle and keep at most 100 edges.
16: for each  $\{i, j\} \in \mathcal{B}_{\text{seed}}$  do
17:   Append  $(x_i^0, x_j^0, q_{ij})$  to  $\mathcal{H}_b^0$ ; increment  $n_b^0$ .
18:   If  $\mathbf{L} = \text{Ridge}$  and  $a_b = \text{single-principal}$ , draw and store  $r_\ell \sim \text{Unif}\{1, 2\}$  for this observation.
19: if  $n_b^0 > 0$  then
20:   Set  $\text{rep}_b^0$  to mean broker seed output.
21: else
22:   Set  $\text{rep}_b^0 \leftarrow 0$ .
23: Set  $s_{i,\text{self}}^0$  to mean own seed output and  $s_{i,\text{broker}}^0 \leftarrow \text{rep}_b^0$ .
24: Set  $\mathbf{p}_i^0 \leftarrow (n_i^0)$  for every agent and  $\mathbf{p}_b^0 \leftarrow (n_b^0)$ , so initialization is period 0; set  $\Delta_i^0 \leftarrow n_i^0$  and  $\Delta_b^0 \leftarrow n_b^0$ .
25: for each  $i$  with  $n_i^0 > 0$  do
26:    $\Theta_i^0 \leftarrow \text{FITPREDICTOR}(\mathbf{L}, a, \Theta_i^0, \Xi_i^0, y_i^0, \lambda_a, n_i^0, n_i^0, \vartheta)$ ;  $\Delta_i^0 \leftarrow 0$ .
27: if  $n_b^0 > 0$  then
28:    $Z_{b,\text{seed}} \leftarrow [z_{b\ell}]_{\ell=1}^{n_b^0}$ ,  $y_{b,\text{seed}} \leftarrow y_b^0$ .
29:    $\Theta_b^0 \leftarrow \text{FITPREDICTOR}(\mathbf{L}, b, \Theta_b^0, Z_{b,\text{seed}}, y_{b,\text{seed}}, \lambda_b, n_b^0, n_b^0, \vartheta)$ ;  $\Delta_b^0 \leftarrow 0$ .
30: return  $S^0$ .

```

5 Channel Choice

Local variables for ChooseChannel

Symbol	Dimension/domain	Definition
U_{self}	scalar	Self-channel score.
U_{broker}	scalar	Broker-channel score.

Algorithm 7 ChooseChannel

Require: Agent i state (s_i, χ_i) , broker reputation rep_b , RNG.

```

1:  $U_{\text{self}} \leftarrow s_{i,\text{self}}$ .
2:  $U_{\text{broker}} \leftarrow \chi_i s_{i,\text{broker}} + (1 - \chi_i) \text{rep}_b$ .
3: if  $U_{\text{self}} > U_{\text{broker}}$  then
4:   return self.
5: if  $U_{\text{broker}} > U_{\text{self}}$  then
6:   return broker.
7: return self w.p.  $1/2$ , else broker  $\triangleright U_{\text{self}} = U_{\text{broker}}$ .

```

6 One Period

Local variables for PeriodUpdate

Symbol	Dimension/domain	Definition
K	integer	Active-demand bound.
p_{demand}	scalar	Per-position demand probability.
α_R	scalar	Standing broker roster share.
p_{roster}	scalar	Roster-churn probability.
η_{exit}	scalar	Agent exit probability.
Δ_{net}	integer	Network-measure interval.

Algorithm 8 PeriodUpdate

Require: Previous state S^{t-1} , period t , parameters ϑ .

Ensure: Updated state S^t , accepted records $\mathcal{R}^t$, and **pre-turnover measure row** Y^t .

- 1: Clear accumulators, set $M_i^t \leftarrow \emptyset$ for all i , and set $D^t \leftarrow \emptyset$.
- 2: Refresh roster: remove each roster member with probability p_{roster} , then replenish uniformly to size $\lceil \alpha_R N \rceil$.
- 3: $\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t\}$.
- 4: $\mathcal{D}^t \leftarrow \emptyset$.
- 5: **for** $i = 1, \dots, N$ **do**
- 6: Draw $d_i^t \sim \text{Binomial}(K, p_{\text{demand}})$.
- 7: **if** $d_i^t > 0$ **then**
- 8: $z_i^t \leftarrow \text{CHOOSECHANNEL}(i, \text{rep}_b^{t-1})$.
- 9: Append (i, z_i^t, d_i^t) to $\mathcal{D}^t$.
- 10: **if** $z_i^t = \text{broker}$ **then**
- 11: Add i to D^t .
- 12: $\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t \cup D^t\}$.
- 13: $\text{TRAINPREDICTORS}(S, t, \vartheta)$.
- 14: $\mathcal{R}^t \leftarrow \text{OFFERMARKET}(S, \mathcal{D}^t, \vartheta)$.
- 15: Synchronize broker-node edges to Roster^t , D^t , and current broker-channel matches.
- 16: $\text{UPDATESATISFACTION}(S, \mathcal{D}^t, \mathcal{R}^t)$.
- 17: **if** $D^t \neq \emptyset$ **then**
- 18: $\text{rep}_b^t \leftarrow |D^t|^{-1} \sum_{i \in D^t} s_{i, \text{broker}}^t$.
- 19: **else**
- 20: $\text{rep}_b^t \leftarrow \text{rep}_b^{t-1}$.
- 21: **if** $t \bmod \Delta_{\text{net}} = 0$ **then**
- 22: **Recompute broker betweenness, Burt constraint, and effective size on the post-matching, pre-turnover graph** G^t .
- 23: $Y^t \leftarrow \text{COLLECTMEASURES}(S^t, \mathcal{R}^t, \vartheta)$ **on the post-matching, pre-turnover state.**
- 24: Append n_i^t to $\mathbf{p}_i^t$ for every agent and append n_b^t to $\mathbf{p}_b^t$, recording the end-of-period training-window boundaries.
- 25: **for** $i = 1, \dots, N$ **do**
- 26: **if** agent i exits with probability η_{exit} **then**
- 27: $\text{INITIALIZEAGENT}(S^t, i, t, \vartheta)$.
- 28: Synchronize broker-node edges to the post-turnover roster, clients, and active matches.
- 29: **return** $(S^t, \mathcal{R}^t, Y^t)$.

7 Offer Market

Local variables for OfferMarket

Symbol	Dimension/domain	Definition
$\hat{q}_i^t$	$\mathbb{R}^d \rightarrow \mathbb{R}$	Agent predictor.
$\hat{q}_b^t$	$\mathbb{R}^{d_b} \rightarrow \mathbb{R}$	Broker predictor.
$\mathcal{N}_G^t(i)$	subset of $[N]$	Non-broker graph neighbors of i .
U_i^t	subset of $[N]$	Self-search stranger pool independently sampled for demander i .
$n_{\text{strangers}}$	integer	Per-demander stranger-pool size.
$v_{i \rightarrow j}^{\text{self}, t}$	scalar	Self-offer score.
$\mathcal{U}_{b,i}$	subset of $[N]$	Candidate set for broker-client demander i .
$v_{b,i \rightarrow j}^t$	scalar	Score used to order brokered candidates for demander i .
Ω_i	integer	Residual offer quota for demander i .
Ω	$\mathbb{Z}_{\geq 0}^N$	Residual quota vector.
O^t	map	Directed offer book.
O_{ij}^t	offer record	Record (i, j, z, v) for offer $i \rightarrow j$.
$\mathcal{P}_O^t$	set	Unordered pairs represented in O^t .
$\mathcal{U}_i^{\text{self}}$	subset of $[N]$	Self-search candidate set.
$\mathcal{I}_i^{\text{self}}$	list	Self-search candidates sorted by score.
$\mathcal{I}_{b,i}$	list	Brokered candidates for client i , sorted by $v_{b,i \rightarrow j}^t$.
$\mathcal{B}_b^t$	list	Selected directed broker offers.
$u_{i \rightarrow j}^{\text{self}}, u_{b,i \rightarrow j}^{\text{local}}, u_{b,i \rightarrow j}^{\text{global}}$	$[0, 1]$	Independent random secondary keys used only within equal-score rankings.
o, o', o_{ij}, o_{ji}	offer records	Offer records; fields include $o.z$ and $o.from$.
z^*	channel	Accepted-record channel.
ι	$\{0, 1\}$	Return value of AddOffer.

The symmetric broker score is

$$v_b^t(i, j) = \text{PREDICTBROKER}(i, j).$$

Let $\mathcal{N}_G^t(i) = \{j \in [N] : (i, j) \in \mathcal{E}_G^t\}$ exclude the broker node. Agent i assesses any presented candidate as

$$a_{i \rightarrow j}^t = \begin{cases} \hat{q}_{ij}^t, & j \in \mathcal{N}_G^t(i), C_{ij}^t > 0, \\ \text{PREDICTAGENT}(i, x_j^t), & \text{otherwise.} \end{cases}$$

Self-search makes this score available only for neighbors and non-neighbors in the demander-specific stranger pool:

$$v_{i \rightarrow j}^{\text{self}, t} = \begin{cases} a_{i \rightarrow j}^t, & j \in \mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i)), \\ \perp, & \text{otherwise.} \end{cases}$$

Thus direct pair means replace predictor extrapolations when pair-specific experience exists; $\perp$ denotes no valid self-search score. The receiver-side acceptance score for a one-sided offer is

$$w_{j \leftarrow i}^t = \begin{cases} \hat{q}_{ji}^t, & (i, j) \in \mathcal{E}_G^t, C_{ji}^t > 0, \\ \text{PREDICTAGENT}(j, x_i^t), & \text{otherwise.} \end{cases}$$

The market creates at most d_i^t outgoing offers from demander i . Receivers have no per-period capacity constraint; acceptance is pairwise, so a receiver may accept multiple offers in the same period. A demanded position can remain unmatched if no acceptable offer is formed.

Algorithm 9 AddOffer

Require: Offer book O^t , unordered-pair set $\mathcal{P}_O^t$, offer (i, j, z, v) .

- 1: **if** O_{ij}^t is defined **then**
 - 2: **return** 0.
 - 3: $O_{ij}^t \leftarrow (i, j, z, v)$.
 - 4: **if** O_{ji}^t is undefined **then**
 - 5: $\mathcal{P}_O^t \leftarrow \mathcal{P}_O^t \cup \{\{i, j\}\}$.
 - 6: **return** 1.
-

Algorithm 10 BrokerCandidates

Require: Broker-client demander i , state S , broker service S_b .

Ensure: Candidate set $\mathcal{U}_{b,i}$ and score function $v_{b,i \rightarrow j}^t$.

- 1: **if** $S_b = \text{full}$ **then**
 - 2: $\mathcal{U}_{b,i} \leftarrow \mathcal{A}_b^t \setminus (\{i\} \cup M_i^t)$ and $v_{b,i \rightarrow j}^t \leftarrow v_b^t(i, j)$.
 - 3: **else if** $S_b = \text{assessment-only}$ **then**
 - 4: Draw $U_i^t \subset [N]$ uniformly without replacement, $|U_i^t| = \min(n_{\text{strangers}}, N)$.
 - 5: $\mathcal{U}_{b,i} \leftarrow [\mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i))] \setminus (\{i\} \cup M_i^t)$ and $v_{b,i \rightarrow j}^t \leftarrow v_b^t(i, j)$.
 - 6: **else**
 - 7: $\mathcal{U}_{b,i} \leftarrow \mathcal{A}_b^t \setminus (\{i\} \cup M_i^t)$ and $v_{b,i \rightarrow j}^t \leftarrow a_{i \rightarrow j}^t$. $\triangleright$ access-only
 - 8: **return** $(\mathcal{U}_{b,i}, v_{b,i \rightarrow j}^t)$.
-

Algorithm 11 OfferMarket

Require: State S , demand records $\mathcal{D}^t$, parameters ϑ .

Ensure: Accepted records $\mathcal{R}^t$.

- 1: For all $(i, z_i, d_i) \in \mathcal{D}^t$, set residual offer quota $\Omega_i \leftarrow d_i$.
- 2: $O^t \leftarrow \emptyset$, $\mathcal{P}_O^t \leftarrow \emptyset$, $\mathcal{R}^t \leftarrow \emptyset$.
- 3: **for** each $(i, \text{self}, d_i) \in \mathcal{D}^t$ with $\Omega_i > 0$ **do**
- 4: Draw $U_i^t \subset [N]$ uniformly without replacement, $|U_i^t| = \min(n_{\text{strangers}}, N)$.
- 5: $\mathcal{U}_i^{\text{self}} \leftarrow \mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i))$.
- 6: Remove from $\mathcal{U}_i^{\text{self}}$ the elements of $\{i\} \cup M_i^t$.
- 7: Draw independent $u_{i \rightarrow j}^{\text{self}} \sim \text{Unif}(0, 1)$ for each feasible j .
- 8: $\mathcal{I}_i^{\text{self}} \leftarrow \text{argsort}_{j \in \mathcal{U}_i^{\text{self}}: v_{i \rightarrow j}^{\text{self}, t} > r}(-v_{i \rightarrow j}^{\text{self}, t}, u_{i \rightarrow j}^{\text{self}})$.
- 9: **for** the first $\min(\Omega_i, |\mathcal{I}_i^{\text{self}}|)$ agents j in $\mathcal{I}_i^{\text{self}}$ **do**
- 10: $\text{ADDOFFER}(O^t, \mathcal{P}_O^t, i, j, \text{self}, v_{i \rightarrow j}^{\text{self}, t})$.
- 11: $\mathcal{A}_b^t \leftarrow \text{Roster}^t \cup D^t$.
- 12: $D_+^t \leftarrow \{i : (i, \text{broker}, d_i) \in \mathcal{D}^t, \Omega_i > 0\}$.
- 13: $\mathcal{B}_b^t \leftarrow \emptyset$.
- 14: **for** each $i \in D_+^t$ **do**
- 15: $(\mathcal{U}_{b,i}, v_{b,i \rightarrow j}^t) \leftarrow \text{BROKERCANDIDATES}(i, S, S_b)$.
- 16: Draw independent $u_{b,i \rightarrow j}^{\text{local}} \sim \text{Unif}(0, 1)$ for each feasible j .
- 17: $\mathcal{I}_{b,i} \leftarrow \text{argsort}_{j \in \mathcal{U}_{b,i}: v_{b,i \rightarrow j}^t > r}(-v_{b,i \rightarrow j}^t, u_{b,i \rightarrow j}^{\text{local}})$.

```

18:   for the first  $\min(\Omega_i, |\mathcal{I}_{b,i}|)$  agents  $j$  in  $\mathcal{I}_{b,i}$  do
19:     Append  $(i, j, v_{b,i \rightarrow j}^t)$  to  $\mathcal{B}_b^t$ .
20: Draw a new independent  $u_{b,i \rightarrow j}^{\text{global}} \sim \text{Unif}(0, 1)$  for each  $(i, j, v) \in \mathcal{B}_b^t$ .
21: Sort  $\mathcal{B}_b^t$  lexicographically by  $(-v, u_{b,i \rightarrow j}^{\text{global}})$ .
22: for each  $(i, j, v) \in \mathcal{B}_b^t$  do
23:   if  $\Omega_i > 0$  then
24:      $\iota \leftarrow \text{ADDOFFER}(O^t, \mathcal{P}_O^t, i, j, \text{broker}, v)$ .
25:      $\Omega_i \leftarrow \Omega_i - \iota$ .
26: for each  $\{i, j\} \in \mathcal{P}_O^t$  do
27:    $o_{ij} \leftarrow O_{ij}^t$  if defined, else  $\emptyset$ ;  $o_{ji} \leftarrow O_{ji}^t$  if defined, else  $\emptyset$ .
28:   if  $o_{ij} \neq \emptyset$  and  $o_{ji} \neq \emptyset$  then
29:      $\text{FINALIZEMATCH}(S, \mathcal{R}^t, o_{ij}, o_{ji})$ .
30:   else if  $o_{ij} \neq \emptyset$  and  $w_{j \leftarrow i}^t > r$  then
31:      $\text{FINALIZEMATCH}(S, \mathcal{R}^t, o_{ij}, \emptyset)$ .
32:   else if  $o_{ji} \neq \emptyset$  and  $w_{i \leftarrow j}^t > r$  then
33:      $\text{FINALIZEMATCH}(S, \mathcal{R}^t, o_{ji}, \emptyset)$ .
34: return  $\mathcal{R}^t$ .

```

Algorithm 12 FinalizeMatch

Require: State S , accepted-record list $\mathcal{R}^t$, offer $o = (i, j, z, v)$, optional reverse offer o' .

```

1: if  $j \in M_i^t$  then
2:   return .
3: Draw  $q_{ij}^t = Q + f(x_i^t, x_j^t) + \varepsilon$ ,  $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ .
4: Append columns  $x_j^t$  to  $\Xi_i$  and  $x_i^t$  to  $\Xi_j$ ; append  $q_{ij}^t$  to  $y_i$  and  $y_j$ ; increment  $n_i$  and  $n_j$ .
5: Update  $\bar{q}_{ij}, C_{ij}$  and  $\bar{q}_{ji}, C_{ji}$ .
6: Add edge  $(i, j)$  to  $G^t$ ; add  $j$  to  $M_i^t$  and  $i$  to  $M_j^t$ .
7: if  $z = \text{broker}$  or  $(o' \neq \emptyset$  and  $o'.z = \text{broker})$  then
8:   Append the two party types  $x_i^t$  and  $x_j^t$  to  $X_{b,1}$  and  $X_{b,2}$ ; append  $q_{ij}^t$  to  $y_b$ ; increment  $n_b$ .
9:   If  $\mathbf{L} = \text{Ridge}$  and  $a_b = \text{single-principal}$ , draw and store  $r_\ell \sim \text{Unif}\{1, 2\}$  for this observation.
10:   $z^* \leftarrow \text{broker}$ .
11: else
12:   $z^* \leftarrow \text{self}$ .
13: Append accepted record  $(i, j, z^*, q_{ij}^t, v, o, o')$  to  $\mathcal{R}^t$ .

```

8 Satisfaction and Measures

Local variables for satisfaction

Symbol	Dimension/domain	Definition
Y_i	scalar	Agent i period satisfaction numerator.
$n_{i,b}$	integer	Accepted broker offers credited to demander i .
$\bar{q}_i^t$	scalar	Channel-specific satisfaction input.
ζ	accepted record	Iterator over $\mathcal{R}^t$.
$\mathbf{1}\{\cdot\}$	indicator	Equals 1 when its condition is true and 0 otherwise.

Algorithm 13 UpdateSatisfaction

Require: State S , demand records $\mathcal{D}^t$, accepted records $\mathcal{R}^t$.

```

1: for each  $(i, z_i, d_i) \in \mathcal{D}^t$  do
2:    $Y_i \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset, o.\text{from} = i\} \zeta.q.$ 
3:    $n_{i,b} \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset, o.\text{from} = i, o.z = \text{broker}\}.$ 
4:   if  $z_i = \text{self}$  then
5:      $\tilde{q}_i^t \leftarrow Y_i/d_i - c_s.$ 
6:   else
7:      $\tilde{q}_i^t \leftarrow (Y_i - \phi n_{i,b})/d_i; \chi_i^t \leftarrow 1.$ 
8:    $s_{i,z_i}^t \leftarrow (1 - \omega)s_{i,z_i}^{t-1} + \omega \tilde{q}_i^t.$ 

```

CollectMeasures. COLLECTMEASURES($S^t, \mathcal{R}^t, \vartheta$) records the measures defined in Appendix B, Section 8, without changing the simulation state.